FA 8 Libres, Ben Alexander

1. (a)

$P(X=0) = \sum P(\text{choosing color C}) \cdot P(X=0 \mid \text{color C})$

- If Red ($N_{red} = 10$, others = 15):

$$\frac{1}{3} \cdot \frac{\binom{17}{4}}{\binom{25}{4}} = \frac{1}{3} \cdot \frac{1365}{12650} = \frac{1365}{37950}$$

- If Black ($N_{Black} = 8$, others = 17):

$$\frac{1}{3} \cdot \frac{\binom{17}{4}}{\binom{25}{4}} = \frac{1}{3} \cdot \frac{2380}{12650} = \frac{2380}{37950}$$

- If Green ($N_{Green} = 7$, others = 18):

$$\frac{1}{3} \cdot \frac{\binom{18}{4}}{\binom{25}{4}} = \frac{1}{3} \cdot \frac{3060}{12650} = \frac{3060}{37950}$$

$$P(X=0) = \frac{1365 + 2380 + 3060}{37950} = \frac{6805}{37950} \approx 0.1793$$

(b). Find $P(X_i = 1)$ for $i = 1, 2, 3, 4$

Let $X_i = 1$ if the ith ball selected is the chosen color, 0 otherwise

$$P(X_i = 1) = \frac{1}{3}\left(\frac{10}{25}\right) + \frac{1}{3}\left(\frac{8}{25}\right) + \frac{1}{3}\left(\frac{7}{25}\right) = \frac{1}{3}\left(\frac{25}{25}\right)$$

$$\approx 0.3333$$

(c) Find $E[X]$

Linearity of Expectation:

$$E[X] = E[X_1] + E[X_2] + E[X_3] + E[X_4]$$

$$= 4\left(\frac{1}{3}\right) = 1.3333$$

2. (a)

Using the hint $\frac{1}{n(n+1)(n+2)} = \frac{1}{2}\left[\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right]$

Then,

$$\sum_{n=1}^{\infty} P(X=n) = 4 \cdot \frac{1}{2} \sum_{n=1}^{\infty} \left(\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right)$$

which is a telescoping series, leaving only the 1st Term.

$$2\left(\frac{1}{1 \cdot 2}\right) = 1$$

(b)

$$E[X] = \sum_{n=1}^{\infty} n \cdot \frac{4}{n(n+1)(n+2)} = \frac{4}{(n+1)(n+2)}$$

$$= 4 \sum_{n=1}^{\infty} \left(\frac{1}{n+1} - \frac{1}{n+2}\right) \Rightarrow \text{a telescoping series}$$

$$4\left(\frac{1}{2} - 0\right) = 2$$

(c)

$$E[X^2] = \sum_{n=1}^{\infty} n^2 \cdot \frac{4}{n(n+1)(n+2)} = \sum_{n=1}^{\infty} \frac{4n}{(n+1)(n+2)}$$

As $n \to \infty$, behaves like $\frac{4}{n}$, a harmonic series which diverges.

Then $E[X^2] = \infty$

3.

$$f(x) = \begin{cases} a + bx^2, & 0 \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$$

$E[X] = 3/5$

- Total Prob = 1, Then $\int_0^1 a + bx^2 \, dx = 1$

$$a + \frac{b}{3} = 1 \rightarrow 3a + b = 3$$

- Expected value: $\int_0^1 x(a + bx^2) \, dx = \frac{3}{5}$

$$\frac{a}{2} + \frac{b}{4} = \frac{3}{5} \rightarrow 10a + 5b = 12$$

Systems of equations:

$3a + b = 3$ since $b = 3 - 3a$

$10a + 5b = 12$ ↓

$10a + 5(3 - 3a) = 12$

$-5a = -3$

$a = \frac{3}{5} = 0.6$

Substituting a

$3\left(\frac{3}{5}\right) + b = 3$

$\frac{9}{5} + b = 3$

$b = 3 - \frac{9}{5} = \frac{15-9}{5} = \frac{6}{5} = 1.2$

4. (a) Expected Value of $Y = \frac{X - \mu}{\sigma}$ where $E[X] = \mu$, $Var(X) = \sigma^2$

$$E[Y] = E\left[\frac{1}{\sigma}X - \frac{\mu}{\sigma}\right] = \frac{1}{\sigma}E[X] - \frac{\mu}{\sigma}$$

$$= \frac{\mu}{\sigma} - \frac{\mu}{\sigma} = 0$$

(b) Variance of Y

$$Var(Y) = Var\left(\frac{1}{\sigma}X - \frac{\mu}{\sigma}\right)$$

$$= \left(\frac{1}{\sigma}\right)^2 Var(X) = \frac{1}{\sigma^2} \cdot \sigma^2 = 1$$

5. $X \sim \lambda_1 = 2$ $Y \sim \lambda_2 = 3$. Let $W = X + Y$.

(a)

$M_W(t) = M_X(t) \cdot M_Y(t)$ since $M_W(t) = e^{(\lambda_1 + \lambda_2)(e^t - 1)}$

$= e^{\lambda_1(e^t-1)} \cdot e^{\lambda_2(e^t-1)} = e^{2(e^t-1)} \cdot e^{3(e^t-1)}$

$= e^{5(e^t-1)}$

(b) By the uniqueness property of MGF,

W corresponds a Poisson Distribution.

Since $W = X + Y$.

W has parameter $\lambda = 5$ ($W \sim Poisson(5)$)

(c)

Mean: $E[W] = \lambda$

$= 5.0000$

2nd Moment: $E[W^2] = Var(W) + E[W]^2$

$= \lambda + \lambda^2$

$= 5 + 25 = 30$